Two sums of integer parts

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The integers a and b are coprime, with $a > b$. Compare the numbers

$$m = \left[\frac{a}{b}\right] + \left[\frac{2a}{b}\right] + \cdots + \left[\frac{(b-1)a}{b}\right]$$

and

$$n = \left[\frac{b}{a}\right] + \left[\frac{2b}{a}\right] + \cdots + \left[\frac{(a-1)b}{a}\right]$$

where $[x]$ denotes the integer part of the real number x.